In a sense, a vector is just an object with some distance (magnitude) and direction to travel in.

Vectors are either written in boldface or with an arrow ($\vec{v}$)

Since vectors have direction, we care about where they start and end.

<u>Initial Point</u> - Starting point of a vector

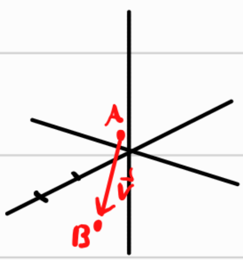
<u>Terminal Point</u> - Ending point of a vector

In fact, just by knowing the initial and terminal point we can recover the vector

<u>Ex:</u> Point $A(1,1,1)$ to $B(2,1,0)$

$$\vec{v} = \overrightarrow{AB} = \langle 1, 0, -1 \rangle$$

Displacement Vector



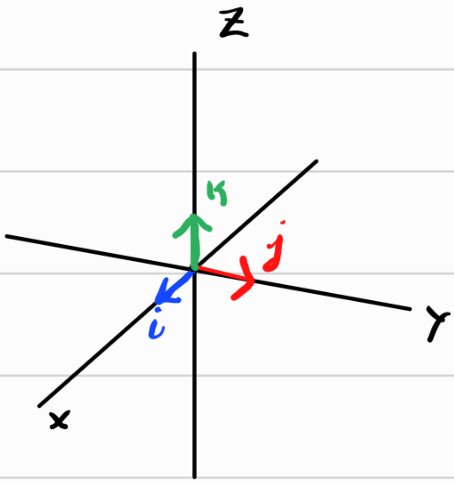
Vectors are often writen as

$$\vec{v} = \langle a, b, c \rangle \quad \text{or} \quad \vec{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Some Special Vectors:

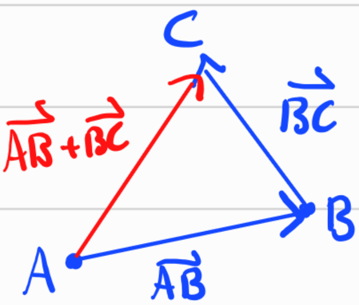
$$\vec{O} = \langle 0, 0, 0 \rangle$$

$$\left.\begin{aligned} \vec{i} &= \langle 1, 0, 0 \rangle \\ \vec{j} &= \langle 0, 1, 0 \rangle \\ \vec{k} &= \langle 0, 0, 1 \rangle \end{aligned}\right\} \text{Standard Basis Vectors}$$



We can add vectors together in a nice way.

If vector $\overrightarrow{AB}$ has initial point A, terminal point B and $\overrightarrow{BC}$ has initial point B, terminal point C then $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ w/ intial point A, terminal point C

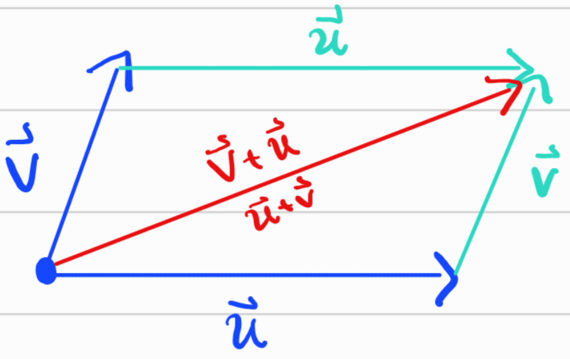


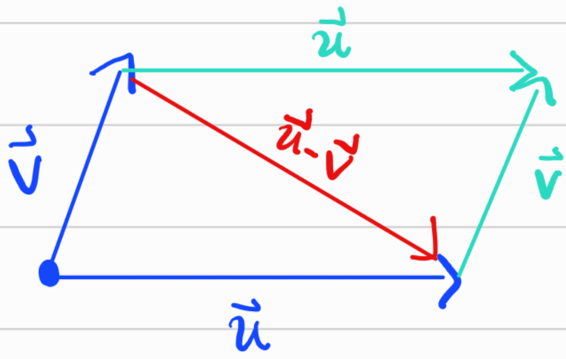
Since we often care about vectors as just direction and magnitude, we often have the initial point be the origin, (or anywhere) this is nice so now all vectors have terminal point as just their components.

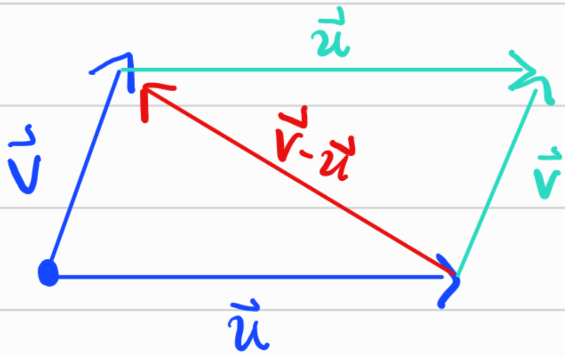
Adding vectors this way is done pointwise

$$\vec{u} = \langle a_1, b_1, c_1 \rangle, \quad \vec{v} = \langle a_2, b_2, c_2 \rangle$$

$$\vec{u} + \vec{v} = \langle a_1 + a_2,\ b_1 + b_2,\ c_1 + c_2 \rangle$$







$$\vec{u} - \vec{v} = -(-\vec{u} + \vec{v}) = -(\vec{v} - \vec{u})$$

In Summary, when adding vectors we are

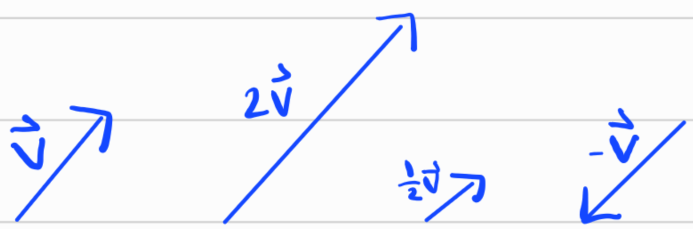
- Geometrically, attaching them end to end
- Algebraically, adding components together

Now, since we have addition one might expect multiplication. However, we can't multiply two vectors. Instead, we can multiply by a scalar (real number)

let $k \in \mathbb{R}$, for $\vec{v} = \langle a, b, c \rangle$

$$k\vec{v} = \langle ka, kb, kc \rangle$$

as the name suggests, we are changing the scale of $\vec{v}$.



Like Calculating distance in $\mathbb{R}^3$, the magnitude of a vector is just its length

for a vector $\vec{v} = \langle a, b, c \rangle$

The magnitude

$$|\vec{v}| = \sqrt{a^2 + b^2 + c^2}$$

Vectors have the following Properties

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$ and $c, d \in \mathbb{R}$ Scalars

1) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$
2) $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$
3) $\vec{a} + \vec{0} = \vec{a}$
4) $\vec{a} + (-\vec{a}) = \vec{0}$
5) $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$
6) $(c+d)(\vec{a}) = c\vec{a} + d\vec{a}$
7) $(cd)\vec{a} = c(d\vec{a})$
8) $1\vec{a} = \vec{a}$

In fact, any Space that Satisfies these properties is called a Vector Space.

Note: There is nothing stopping us from considering n-dimensional vectors, all our definitions and properties still work in any dimension.

Recall the standard basis vectors from earlier

$$\vec{i} = \langle 1,0,0 \rangle, \quad \vec{j} = \langle 0,1,0 \rangle, \quad \vec{k} = \langle 0,0,1 \rangle$$

Using our rules, we can write any other vector as a linear combination of these three,

Say $\vec{v} = \langle a,b,c \rangle$

then $\vec{v} = a\vec{i} + b\vec{j} + c\vec{k}$

both presentations of $\vec{v}$ are common, so understand both

A <u>Unit vector</u> is a vector $\vec{v}$ such that $|\vec{v}| = 1$

Given any vector $\vec{a}$, we can find a unit vector in the same direction as $\vec{a}$ by simply dividing by $|\vec{a}|$, that is

$\vec{u} = \frac{1}{|\vec{a}|}\vec{a}$ is a unit vector in the direction of $\vec{a}$.